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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Examinations August-2022
Common to CSE/ISE
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

*Time: 03 Hours**Maximum Marks: 100***Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	An _____ is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through_____.	02
	1.2	Write LMS weight update rule.	02
	1.3	List the issues in Machine Learning.	02
	1.4	_____ is the heuristic used in ID3 algorithm for selecting node for growing the tree.	01
	1.5	Mention the alternative measures for selecting attributes in decision tree learning.	02
	1.6	Beam-search with _____ beam-width is called as hill-climbing problem.	01
	1.7	Mention the two approaches that are used for avoiding overfitting in decision tree learning.	02
	1.8	List the key differences between lazy and eager methods in the inductive bias	02
	1.9	Radial basis function (RBF) networks are a type of artificial neural network constructed from _____ functions	01
	1.10	Write PEAS description for Playing a tennis match.	02
	1.11	The number of states in a vacuum world problem with 5 states is _____	01
	1.12	Why does A* algorithm consumes more memory than the greedy BFS?	01
	1.13	Which kind of agent takes action ignoring the percept history?	01

PART B

2	a	Explain uniform cost search algorithm with an example.	08
	b	Discuss memory bounded heuristic search in detail.	08
3	a	Use the diagram given in Fig. 3a to solve the map colouring problem as a CSP. Domain is given as {red, yellow, blue}. Use proper notations to show variables, constraints and two possible solutions to this problem. Represent this problem as a constraint graph too.	

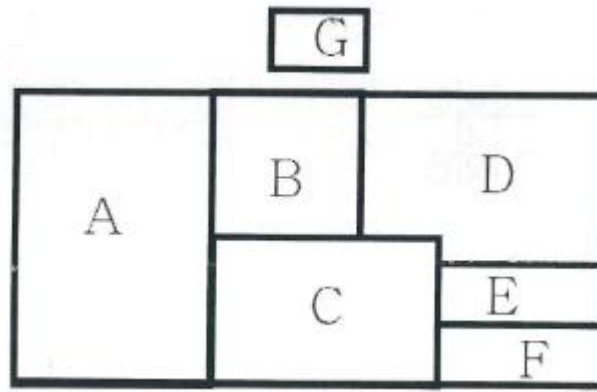


Fig. 3a

08

- b Apply alpha-beta pruning method for the game tree shown in Fig. 3b and find the value at the root also mention the nodes which are pruned.

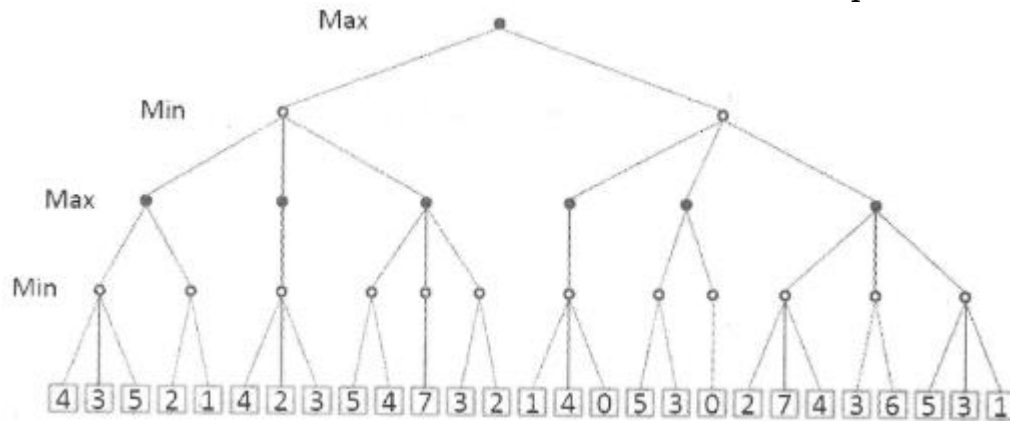


Fig. 3b

08

OR

- 4 a Consider the following sentences:

1. Marcus was a man
2. Marcus was a Pompeian
3. All Pompeians were Romans
4. Caesar was a ruler
5. All Romans were either loyal to Caesar or hated him
6. Everyone is loyal to someone
7. People only try to assassinate rulers they aren't loyal to.
8. Marcus tried to assassinate Caesar

- i. Translate these sentences into formulas in predicate logic. Convert the formulas of Part i into clause form.
- ii. Use Resolution method to answer the question "Marcus hates Caesar."

09

- b Consider the following three-dimensional tic-tac-toe in Fig. 4b and apply Mini-Max algorithm

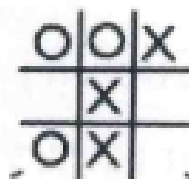


Fig. 4b

07

5	a	Given the Bayesian network shown in Fig. 5a, compute the following probabilities: 1. $P(B=\text{good}, F=\text{empty}, G=\text{empty}, S=\text{yes})$ 2. $P(B=\text{bad}, F=\text{empty}, G=\text{not empty}, S=\text{no})$	
		Fig. 5a	08
	b	Explain the following with an example: i. Semantic networks ii. Description logic.	08
		OR	
6	a	Design semantic nets for the following: i. Every dog has bitten a shopkeeper ii. Every dog in town has bitten a shopkeeper.	08
	b	Explain the enumeration algorithm for answering queries on Bayesian network with an example.	08
7	a	Write short note on i. Restriction bias ii. Preference bias	08
	b	Explain the techniques that can be used for avoiding Overfitting the data problem in decision tree learning.	08
8	a	Explain distance weighted nearest weighted algorithm in detail.	08
	b	Explain Q-learning algorithm with an example.	08